

Design a Unique ID Generator - Step 1: Understand the Problem & Establish Design Scope

1. Requirement Clarification (Interview Discussion)

Candidate–Interviewer Interaction

Candidate: What are the characteristics of unique IDs?

Interviewer: IDs must be **unique and sortable**.

Candidate: Do IDs increment by 1?

Interviewer: They increase over time, but **not necessarily by 1**.

Candidate: Do IDs contain only numerical values?

Interviewer: Yes.

Candidate: What is the ID size requirement?

Interviewer: IDs must fit in **64-bit**.

Candidate: What is the scale of the system?

Interviewer: Must generate **10,000 IDs per second**.

2. Final Requirements Summary

Functional Requirements

- IDs must be **unique**
- IDs must be **sortable (time-ordered)**
- IDs must be **numeric only**

Non-Functional Requirements

- Must fit into **64-bit**
- Must support **high throughput (10,000+ IDs/sec)**

3. Key Takeaway

Step 1 focuses on:

- Understanding constraints
- Clarifying requirements
- Defining system scale